

TM4C123 Beam Balancing Dashboard – Setup Guide

This document explains the required installations and setup steps to successfully run the Python Tkinter + Serial + Matplotlib dashboard for TM4C123 beam balancing telemetry.

1. Install Python

- Download Python 3.9–3.11 from <https://www.python.org/downloads/>
- During installation, check 'Add Python to PATH'
- Verify installation using:
`python --version`

2. Required Python Libraries

The dashboard requires the following libraries:

- pyserial – for UART communication
- matplotlib – for real-time plotting
- tkinter – for GUI (included with Python on Windows)

3. Install Required Libraries

Open Command Prompt or PowerShell and run:
`pip install pyserial matplotlib`

4. COM Port Configuration

- Open Device Manager → Ports (COM & LPT)
- Identify the COM port for TM4C123
- Update the code accordingly:
`PORT = "COM3"`

5. UART Data Format (Microcontroller Side)

The TM4C123 must transmit data in the following exact format:

RPM1:1234 RPM2:1250 I1:1.25 I2:1.30 Roll:-2.45

Keywords are case-sensitive and must include:
RPM1, RPM2, I1, I2, Roll

6. Running the Program

- Save the Python file as `dashboard.py`
- Run using:
`python dashboard.py`

7. Common Errors and Fixes

- ModuleNotFoundError: serial → Install pyserial
- No data shown → Check COM port and UART format
- COM port access denied → Close other serial applications

8. Final Checklist

- ✓ Python installed
- ✓ Required libraries installed
- ✓ Correct COM port selected
- ✓ Correct UART output format
- ✓ No other application using COM port